

ENME608 Mechanisms

Machinery Dynamics — Fundamentals (Norton Ch. 10)

Aaron Taylor — 24232594

2026

Contents

1	Newton's Laws of Particle Motion	2
2	Inertial Parameters	2
2.1	Mass	2
2.2	Centre of Mass / Centre of Gravity (COM/COG)	2
2.3	Moment of Inertia (MOI)	2
2.4	Parallel Axis Theorem	3
2.5	Radius of Gyration	3
3	MOI of Standard Homogeneous Shapes	3
4	Moment of Force (Torque)	4
5	Newton's 2nd Law — Rotational Motion	4
6	Measurement of MOI (Physical Pendulum)	4
7	Work, Energy, and Power	5
8	D'Alembert's Principle	5
9	Energy Methods — Virtual Work	6
10	Equivalent (Energy-Wise) Systems via Lever/Gear Ratios	6
11	Unit Systems — IPS vs. SI	6
12	Tutorial Solutions	7
12.1	Problem 10-1 — Composite Mallet: CG, MOI, and Radius of Gyration	7
12.2	Problem 10-4 — System of Equations in Matrix Form	9
12.3	Problem 10-22 — Effective Inertia Reflected Through Winch Drive	9
12.4	Problem 10-24 — Equivalent Mass and Spring via Lever	10

Newton's Laws of Particle Motion

- **Law 1 — Inertia:** A body at rest or moving at constant velocity remains so unless acted on by a net external force.
- **Law 2 — $F = ma$:**

$$\mathbf{F} = m\mathbf{a} = \frac{d\mathbf{M}}{dt}, \quad \mathbf{M} = m\mathbf{v} \quad (\text{linear momentum})$$

- **Law 3 — Action–Reaction:**

$$\mathbf{F}_1 = -\mathbf{F}_2$$

Inertial Parameters

Mass

$$m = \int dm$$

Centre of Mass / Centre of Gravity (COM/COG)

For a continuous body:

$$\bar{X} = \frac{\int x \, dm}{\int dm}, \quad \bar{Y} = \frac{\int y \, dm}{\int dm}, \quad \bar{Z} = \frac{\int z \, dm}{\int dm}$$

For a **composite body** (sum of discrete parts):

$$\bar{X} = \frac{\sum_i m_i x_i}{\sum_i m_i}$$

Exam focus. Exam technique: Replace m_i with weight $W_i = \gamma_i V_i$ (weight density $\times$ volume) since g cancels in numerator and denominator:

$$\bar{X} = \frac{\sum W_i x_i}{\sum W_i}$$

This avoids the $g = 386 \text{ in/s}^2$ conversion until you need actual mass for MOI.

Composite example (handle length l_h , disk head):

$$d = \frac{m_h \frac{l_h}{2} + m_d l_d}{m_h + m_d}$$

where l_d = distance from reference axis to head CG.

Moment of Inertia (MOI)

$$I_x = \int (y^2 + z^2) \, dm, \quad I_y = \int (x^2 + z^2) \, dm, \quad I_z = \int (x^2 + y^2) \, dm$$

Parallel Axis Theorem

Relates MOI about any axis to the MOI about the *parallel* axis through the COM:

$$I_{ZZ} = I_{GG} + md^2$$

where d = perpendicular distance between the two parallel axes.

For the composite handle + disk example:

$$I_{ZZ} = \underbrace{\left[I_{HH} + m_h \left(\frac{l_h}{2} \right)^2 \right]}_{\text{handle contribution}} + \underbrace{[I_{DD} + m_d l_d^2]}_{\text{head contribution}}$$

Standard centroidal MOI for a solid cylinder (transverse axis through its own CG):

$$I_{HH} = \frac{m_h(3r_h^2 + l_h^2)}{12}, \quad I_{DD} = \frac{m_d(3r_d^2 + h_d^2)}{12}$$

Radius of Gyration

The equivalent radius at which the entire mass would give the same MOI:

$$I_{ZZ} = mk^2 \implies k = \sqrt{\frac{I_{ZZ}}{m}}$$

Combined with the parallel axis theorem:

$$I_{GG} + md^2 = mk^2 \implies k^2 = k_{GG}^2 + d^2 > d^2$$

Exam focus. $k > d$ **always.** The radius of gyration is always greater than the distance from the reference axis to the COM. This provides a useful sanity check.

MOI of Standard Homogeneous Shapes

All formulas below are about axes through the **COM**. Use the parallel axis theorem to shift to any other parallel axis.

Shape	Transverse axis ($I_{xx} = I_{yy}$)	Spin axis (I_{zz})
Sphere (radius r)	$\frac{2}{5}mr^2$	$\frac{2}{5}mr^2$
Hemisphere	$0.259mr^2$	$\frac{2}{5}mr^2$
Thin circular disk	$\frac{1}{4}mr^2$	$\frac{1}{2}mr^2$
Thin ring	$\frac{1}{2}mr^2$	mr^2
Solid cylinder (r, h)	$\frac{m(3r^2 + h^2)}{12}$	$\frac{1}{2}mr^2$
Cone (r, h)	$\frac{3m(4r^2 + h^2)}{80}$	$\frac{3}{10}mr^2$
Thin plate ($a \times b$)	$I_{xx} = \frac{1}{12}mb^2, I_{yy} = \frac{1}{12}ma^2$	$\frac{m(a^2 + b^2)}{12}$
Slender rod (length l)	$\frac{1}{12}ml^2$ (mid) / $\frac{1}{3}ml^2$ (end)	0

Exam focus. Rod about its end: Comes directly from the parallel axis theorem: $\frac{1}{12}ml^2 + m(\frac{l}{2})^2 = \frac{1}{3}ml^2$. Memorise this — it appears in most composite-body problems.

Moment of Force (Torque)

As a vector cross product:

$$\mathbf{M}_O = \mathbf{r}_{OA} \times \mathbf{F}$$

Lecture example ($\mathbf{F}_1 = \{100\mathbf{i} - 120\mathbf{j} + 75\mathbf{k}\}$ kN, $\mathbf{F}_2 = \{-200\mathbf{i} + 250\mathbf{j} + 100\mathbf{k}\}$ kN, $\mathbf{r}_{OA} = \{4\mathbf{i} + 5\mathbf{j} + 3\mathbf{k}\}$ m):

$$\mathbf{M}_O = \mathbf{r}_{OA} \times (\mathbf{F}_1 + \mathbf{F}_2) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 5 & 3 \\ -100 & 130 & 175 \end{vmatrix} = \{485\mathbf{i} - 1000\mathbf{j} + 1020\mathbf{k}\} \text{ kNm}$$

Newton's 2nd Law — Rotational Motion

$$T = I\alpha$$

T = resultant torque *about the mass centre*; I = MOI about the axis through the COM; α = angular acceleration.

This is the rotational analogue of $F = ma$.

Measurement of MOI (Physical Pendulum)

Suspend the body from a pivot Z at distance r from COM and measure the oscillation period τ .

Equation of motion:

$$-(W \sin \theta) r = I_{ZZ} \ddot{\theta}$$

For small angles ($\sin \theta \approx \theta$) this is simple harmonic motion (SHM). Rearranging:

$$\ddot{\theta} = -\frac{Wr}{I_{ZZ}} \theta \implies \omega_n = \sqrt{\frac{Wr}{I_{ZZ}}} \implies \tau = 2\pi \sqrt{\frac{I_{ZZ}}{Wr}}$$

Solving for I_{ZZ} :

$$I_{ZZ} = Wr \left(\frac{\tau}{2\pi} \right)^2$$

Apply the parallel axis theorem to find the centroidal MOI:

$$I_{GG} = Wr \left(\frac{\tau}{2\pi} \right)^2 - \frac{W}{g} r^2$$

Exam focus. In IPS: W in lb, r in in, τ in s $\Rightarrow I_{ZZ}$ in lb·in·s². Note $W/g = m$ (mass in blobs). Don't forget to subtract the mr^2 shift term to recover I_{GG} .

Work, Energy, and Power

• **Work:** $W_{\text{work}} = \mathbf{F} \cdot \mathbf{R}$ (force dot displacement)

• **Power:** $P = \frac{dE}{dt}$

• **Kinetic energy:**

$$KE = \frac{1}{2}mv^2 \text{ (translational)} + \frac{1}{2}I\omega^2 \text{ (rotational)}$$

• **Potential energy:**

$$PE = mgh \text{ (gravitational)} + \frac{1}{2}kx^2 \text{ (spring)}$$

D'Alembert's Principle

Add an **inertia force** ($-m\mathbf{a}$) and **inertia torque** ($-I\alpha$) to convert a dynamic problem into an equivalent *static* equilibrium:

$$\sum \mathbf{F} - m\mathbf{a} = \mathbf{0} \quad (\text{inertia force})$$

$$\sum T - I\alpha = 0 \quad (\text{inertia torque})$$

Centripetal vs. Centrifugal Forces:

For a mass m rotating at radius r with angular velocity ω :

• **Centripetal force** $= mr\omega^2$ (directed *inward*) — the real force that constrains the mass to circular motion.

• **Centrifugal force** $= mr\omega^2$ (directed *outward*) — this is the D'Alembert inertia force. It is *fictitious*: it is the reaction force on the *constraint*, not a force on the moving body.

Exam focus. D'Alembert lets you write $\sum F = 0$ and $\sum T = 0$ (statics style) for a dynamic problem, provided you add the inertia force/torque terms with a *negative* sign.

Energy Methods — Virtual Work

For a linkage of n links (link 1 is ground), rate of external energy supply equals rate of energy stored within the system (no losses):

$$\sum_{k=2}^n \mathbf{F}_k \cdot \mathbf{v}_k + \sum_{k=2}^n T_k \cdot \omega_k = \sum_{k=2}^n m_k \mathbf{a}_k \cdot \mathbf{v}_k + \sum_{k=2}^n I_k \alpha_k \cdot \omega_k$$

Rearranged to the **virtual work equation** (set to zero):

$$\sum F_k v_k + \sum T_k \omega_k - \sum m_k a_k v_k - \sum I_k \alpha_k \omega_k = 0$$

Exam focus. This is a **scalar** equation — no need to decompose vectors into components. Once the kinematics (all v_k , a_k , ω_k , α_k) are known, one unknown force or torque can be solved directly. Extremely powerful for mechanism analysis.

Equivalent (Energy-Wise) Systems via Lever/Gear Ratios

Consider a lever with pivot between points A (arm length a) and B (arm length b). Lever geometry gives:

$$v_A = \frac{a}{b} v_B \implies \frac{v_B}{v_A} = \frac{b}{a}$$

Effective mass at A equivalent to m_B at B (equal kinetic energies):

$$\frac{1}{2} m_{\text{eff}} v_A^2 = \frac{1}{2} m_B v_B^2 \implies m_{\text{eff}} = \left(\frac{b}{a} \right)^2 m_B$$

Effective spring at B equivalent to k_A at A (equal potential energies):

$$\frac{1}{2} k_{\text{eff}} x_B^2 = \frac{1}{2} k_A x_A^2 \implies k_{\text{eff}} = \left(\frac{a}{b} \right)^2 k_A$$

Effective moment of inertia of a load reflected through a gear train to a reference shaft:

$$I_{\text{eff}} = I_{\text{load}} \left(\frac{\omega_{\text{load}}}{\omega_{\text{ref}}} \right)^2$$

Exam focus. Rule of thumb: Reflecting a load *through a speed increase* makes the load appear *heavier* (larger effective inertia or mass) at the slower shaft. Reflecting through a speed *decrease* makes it appear lighter.

Unit Systems — IPS vs. SI

Exam focus. Default in this course: IPS (inch–pound–second) unless otherwise stated. The IPS unit of mass is the **blob** ($= \text{lb} \cdot \text{s}^2 / \text{in}$). In IPS: $g = 386 \text{ in/s}^2$, so $m = W/386 \text{ blob}$.

Quantity	Symbol	IPS unit	SI unit
Force	F	pound (lb)	newton (N)
Length	l	inch (in)	metre (m)
Time	t	second (s)	second (s)
Mass	m	blob = lb·s ² /in	kilogram (kg)
Weight	W	pound (lb)	newton (N)
Torque	T	lb·in	N·m
MOI	I	lb·in·s ²	kg·m ²
Energy	E	in·lb	joule (J)

Key conversions: 1 blob = 175.127 kg; 1 lb = 4.448 N; 1 in = 0.0254 m.

Tutorial Solutions

Problem 10-1 — Composite Mallet: CG, MOI, and Radius of Gyration

^{*†}10-1 The mallet shown in Figure 10-2 has the following specifications: The steel head has a 1.25-in diameter and is 3.5 in tall; the wood handle is 1.5-in diameter and 10 in long and necks down to 3/4 in wide where it enters the head. Find the location of its composite CG, and its moment of inertia and radius of gyration about axis ZZ . Assume the wood has a density equal to 0.9 times that of water.

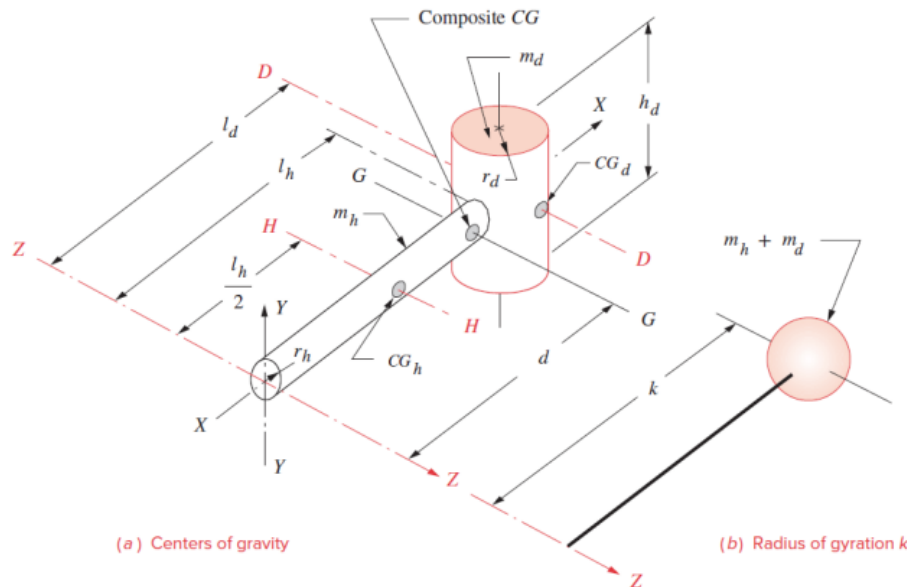


FIGURE 10-2

Problem 10-1: composite mallet statement and figure.

Given:

- Brass head (solid cylinder): $D_d = 1.25$ in $\Rightarrow r_d = 0.625$ in, $h_d = 3.5$ in, $\gamma_d = 0.31$ lbf/in³
- Wood handle (solid cylinder): $D_h = 1.5$ in $\Rightarrow r_h = 0.75$ in, $l_h = 10$ in, $\gamma_h = 0.9 \times 0.036 = 0.0324$ lbf/in³
- Reference axis ZZ passes through the grip end of the handle.

- Head is centred at the end of the handle ($x_d = l_h = 10$ in from ZZ).

Step 1 — Volumes and weights:

$$\begin{aligned} V_h &= \pi r_h^2 l_h = \pi(0.75)^2(10) = 17.671 \text{ in}^3 & W_h &= \gamma_h V_h = 0.0324 \times 17.671 = 0.573 \text{ lbf} \\ V_d &= \pi r_d^2 h_d = \pi(0.625)^2(3.5) = 4.295 \text{ in}^3 & W_d &= \gamma_d V_d = 0.31 \times 4.295 = 1.331 \text{ lbf} \end{aligned}$$

Step 2 — Masses (IPS, $g = 386 \text{ in/s}^2$):

$$m_h = \frac{W_h}{g} = \frac{0.573}{386} = 1.484 \times 10^{-3} \text{ bl}, \quad m_d = \frac{W_d}{g} = \frac{1.331}{386} = 3.449 \times 10^{-3} \text{ bl}$$

Step 3 — CG positions from ZZ :

$$\begin{aligned} x_h &= \frac{l_h}{2} = 5.0 \text{ in} && (\text{handle CG at midpoint}) \\ x_d &= l_h = 10.0 \text{ in} && (\text{head centred at end of handle}) \end{aligned}$$

Step 4 — Composite CG:

$$\bar{X}_{CG} = \frac{W_h x_h + W_d x_d}{W_h + W_d} = \frac{0.573(5.0) + 1.331(10.0)}{0.573 + 1.331} = \frac{2.865 + 13.31}{1.904} = \boxed{8.774 \text{ in}}$$

Step 5 — Centroidal MOI of each part (transverse axis through own CG):

$$\begin{aligned} I_{GG,h} &= \frac{m_h(3r_h^2 + l_h^2)}{12} = \frac{1.484 \times 10^{-3}(3(0.75)^2 + (10)^2)}{12} \\ &= \frac{1.484 \times 10^{-3}(101.69)}{12} = 1.257 \times 10^{-2} \text{ lb} \cdot \text{in} \cdot \text{s}^2 \\ I_{GG,d} &= \frac{m_d(3r_d^2 + h_d^2)}{12} = \frac{3.449 \times 10^{-3}(3(0.625)^2 + (3.5)^2)}{12} \\ &= \frac{3.449 \times 10^{-3}(13.422)}{12} = 3.857 \times 10^{-3} \text{ lb} \cdot \text{in} \cdot \text{s}^2 \end{aligned}$$

Step 6 — Parallel axis theorem to obtain I_{ZZ} :

$$\begin{aligned} I_{ZZ,h} &= I_{GG,h} + m_h x_h^2 = 1.257 \times 10^{-2} + 1.484 \times 10^{-3}(5.0)^2 \\ &= 1.257 \times 10^{-2} + 3.710 \times 10^{-2} = 4.967 \times 10^{-2} \text{ lb} \cdot \text{in} \cdot \text{s}^2 \\ I_{ZZ,d} &= I_{GG,d} + m_d x_d^2 = 3.857 \times 10^{-3} + 3.449 \times 10^{-3}(10.0)^2 \\ &= 3.857 \times 10^{-3} + 3.449 \times 10^{-1} = 3.488 \times 10^{-1} \text{ lb} \cdot \text{in} \cdot \text{s}^2 \\ I_{ZZ} &= I_{ZZ,h} + I_{ZZ,d} = 0.04967 + 0.3488 = \boxed{0.394 \text{ lb} \cdot \text{in} \cdot \text{s}^2} \end{aligned}$$

Step 7 — Radius of gyration:

$$\begin{aligned} m_{\text{total}} &= m_h + m_d = (1.484 + 3.449) \times 10^{-3} = 4.933 \times 10^{-3} \text{ bl} \\ k &= \sqrt{\frac{I_{ZZ}}{m_{\text{total}}}} = \sqrt{\frac{0.394}{4.933 \times 10^{-3}}} = \boxed{9.354 \text{ in}} \end{aligned}$$

Summary of answers:

$$\bar{X}_{CG} = 8.774 \text{ in}, \quad I_{ZZ} = 0.394 \text{ lb} \cdot \text{in} \cdot \text{s}^2, \quad k = 9.354 \text{ in}$$

Sanity check: $k = 9.354 \text{ in} > \bar{X}_{CG} = 8.774 \text{ in} \checkmark$ (radius of gyration exceeds COM distance)

Problem 10-4 — System of Equations in Matrix Form

Set up $[A]\{x\} = \{b\}$ for each system. (Numerical solution via calculator/MATLAB.)

Part (a):

$$\begin{cases} -5x - 2y + 12z - w = -9 \\ x + 3y - 2z + 4w = 10 \\ -x - y + z + 0w = -7 \\ 3x - 3y + 7z + 9w = -6 \end{cases} \Rightarrow \begin{bmatrix} -5 & -2 & 12 & -1 \\ 1 & 3 & -2 & 4 \\ -1 & -1 & 1 & 0 \\ 3 & -3 & 7 & 9 \end{bmatrix} \begin{Bmatrix} x \\ y \\ z \\ w \end{Bmatrix} = \begin{Bmatrix} -9 \\ 10 \\ -7 \\ -6 \end{Bmatrix}$$

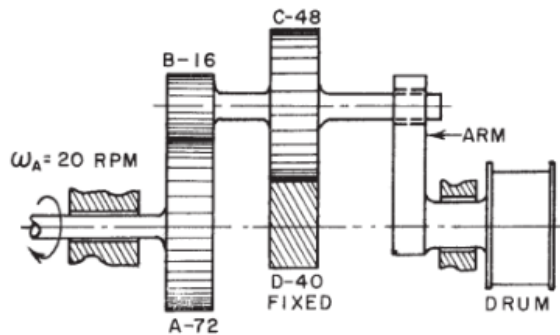
Part (b):

$$\begin{cases} 3x - 5y + 17z - 5w = -5 \\ -2x + 9y - 14z + 6w = 22 \\ -x - y + 0z - 2w = 13 \\ 4x - 7y + 8z + 4w = -9 \end{cases} \Rightarrow \begin{bmatrix} 3 & -5 & 17 & -5 \\ -2 & 9 & -14 & 6 \\ -1 & -1 & 0 & -2 \\ 4 & -7 & 8 & 4 \end{bmatrix} \begin{Bmatrix} x \\ y \\ z \\ w \end{Bmatrix} = \begin{Bmatrix} -5 \\ 22 \\ 13 \\ -9 \end{Bmatrix}$$

Exam focus. The exam will test whether you can correctly map coefficients into the matrix. Key rule: zero coefficients (missing terms) must appear as explicit zeros in $[A]$.

Problem 10-22 — Effective Inertia Reflected Through Winch Drive

†10-22 What is the effective inertia of a load applied at the drum of Figure P9-5a as reflected back to gear A?



(a)

FIGURE P9-5

Problem 10-22 statement and figure.

Context: Figure P9-5a shows the compound planetary winch drive analysed in Tutorial 8 Problem 9-35 (A-72, B-16, C-48, D-40 fixed, ARM = output/drum).

From Tutorial 8 (P9-35):

$$\omega_A = 18 \text{ rpm (input)}, \quad \omega_{\text{drum}} = 22.09 \text{ rpm (same direction)}$$

Velocity ratio:

$$\frac{\omega_{\text{drum}}}{\omega_A} = \frac{22.09}{18} = 1.2272$$

Effective inertia of the drum load as seen at gear A:

$$I_{\text{eff},A} = I_{\text{drum}} \left(\frac{\omega_{\text{drum}}}{\omega_A} \right)^2 = I_{\text{drum}} (1.2272)^2 = \boxed{1.506 I_{\text{drum}}}$$

Exam focus. The drum rotates *faster* than gear A, so the drum load appears *heavier* (larger effective inertia) when viewed from gear A. Formula: $I_{\text{eff}} = I_{\text{load}} (\omega_{\text{load}} / \omega_{\text{ref}})^2$. This follows directly from equating kinetic energies: $\frac{1}{2} I_{\text{eff}} \omega_A^2 = \frac{1}{2} I_{\text{drum}} \omega_{\text{drum}}^2$.

Problem 10-24 — Equivalent Mass and Spring via Lever

†10-24 Refer to Figure 10-10. Given $a = 100$ mm, $b = 150$ mm, $k_A = 2000$ N/m, and $m_B = 2$ kg, find the equivalent mass at point A and the equivalent spring at point B.

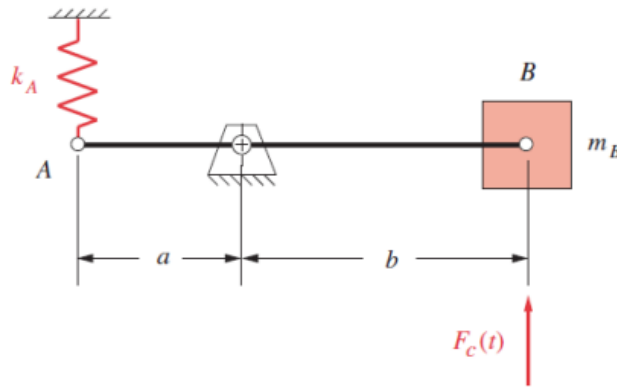


FIGURE 10-10

Problem 10-24 statement and figure.

Given (Figure 10-10, lever with fulcrum between A and B):

$$a = 100 \text{ mm}, \quad b = 150 \text{ mm}, \quad k_A = 2000 \text{ N/m (spring at A)}, \quad m_B = 2 \text{ kg (mass at B)}$$

Velocity relationship from lever geometry:

$$\frac{v_A}{v_B} = \frac{a}{b} = \frac{100}{150} = \frac{2}{3} \quad \Rightarrow \quad v_B = \frac{b}{a} v_A = 1.5 v_A$$

Effective mass at A (equal kinetic energies):

$$\frac{1}{2} m_{\text{eff}} v_A^2 = \frac{1}{2} m_B v_B^2 \quad \Rightarrow \quad m_{\text{eff}} = m_B \left(\frac{b}{a} \right)^2 = 2 \left(\frac{150}{100} \right)^2 = 2 \times 2.25 = \boxed{4.5 \text{ kg at point A}}$$

Effective spring at B (equal potential energies; $x_A = (a/b) x_B$):

$$\frac{1}{2} k_{\text{eff}} x_B^2 = \frac{1}{2} k_A x_A^2 \quad \Rightarrow \quad k_{\text{eff}} = k_A \left(\frac{a}{b} \right)^2 = 2000 \left(\frac{100}{150} \right)^2 = 2000 \times 0.44 = \boxed{888.9 \text{ N/m at point B}}$$

Interpretation: The spring $k_A = 2000$ N/m at the *short* arm appears softer ($k_{\text{eff}} = 888.9$ N/m) when referred to the *long* arm at B. The mass $m_B = 2$ kg at the *long* arm appears heavier ($m_{\text{eff}} = 4.5$ kg) when referred to the *short* arm at A.